INTRODUCTION TO APACHE TOMCAT & BASIC WEB APPLICATION

1. Aim

To install and configure Apache Tomcat and create and run a simple web application using an HTML page.

2. Introduction to Web Applications

A web application is an application that can be accessed through a web browser using a URL. In this experiment: Student → Web Browser → Apache Tomcat → MyWebApp → index.html.

3. What is Apache Tomcat?

Apache Tomcat is a web server and Servlet container used to run Java web applications. For this experiment, Tomcat serves our static HTML page. Later it can be used to run Java Servlets.

4. Why Do We Need Tomcat?

A normal Java program is executed using javac and java. A web application is accessed through a browser using a URL, so a web server is needed to receive browser requests. Tomcat provides this server environment.

5. Tomcat Directory Structure

apache-tomcat-11.0.25
│
├── bin
├── conf
├── lib
├── logs
├── temp
├── webapps
├── work
└── ...

- bin — scripts used to start and stop Tomcat, such as startup.bat and shutdown.bat.
- conf — Tomcat configuration files.
- logs — Tomcat log files.

· lib — libraries used by Tomcat.

· webapps — default deployment directory for web applications.

6. What is webapps?

webapps stands for web applications. It is Tomcat's default deployment directory. Each web application can have its own folder.

7. Creating Our First Web Application

Inside Tomcat's webapps folder, create a folder named MyWebApp.

Complete path:
apache-tomcat-11.0.25\webapps\MyWebApp

8. Creating index.html

Folder structure:

```
webapps
└── MyWebApp
    └── index.html
```

Example index.html:

```
<!DOCTYPE html>
<html>
<head>
   <title>My Web Application</title>
</head>
<body>
   <h1>Welcome to My Web Application</h1>
   <p>This application is running on Apache Tomcat.</p>
</body>
</html>
```

9. What is index.html?

index.html is an HTML page that can act as the default or welcome page of our web application. When the application URL is opened, Tomcat can serve MyWebApp/index.html.

10. Starting Tomcat

Open Tomcat's bin directory and start Tomcat using startup.bat. On Windows PowerShell, use .\startup.bat.

11. What Happens When We Start Tomcat?

Tomcat starts its server and listens for incoming HTTP requests. In this setup, Tomcat uses port 8080.

12. Understanding localhost

localhost means this computer itself. It refers to the machine on which Tomcat is running.

13. Understanding Port 8080

A port number identifies a network service/application. In this experiment Tomcat uses port 8080, so localhost:8080 refers to the Tomcat server running on this computer.

14. Understanding the URL

http://localhost:8080/MyWebApp/

http:// → Protocol
localhost → Computer/server
:8080 → Port
/MyWebApp/ → Web application / context path

This URL means: use HTTP to access the Tomcat server running on this computer at port 8080 and request the MyWebApp application.

15. URL Generation / URL Structure

General formula:
http://<server>:<port>/<application-name>/

For our application:
http://localhost:8080/MyWebApp/

The MyWebApp folder name becomes the application path used in the URL.

16. Relationship Between Folder and URL

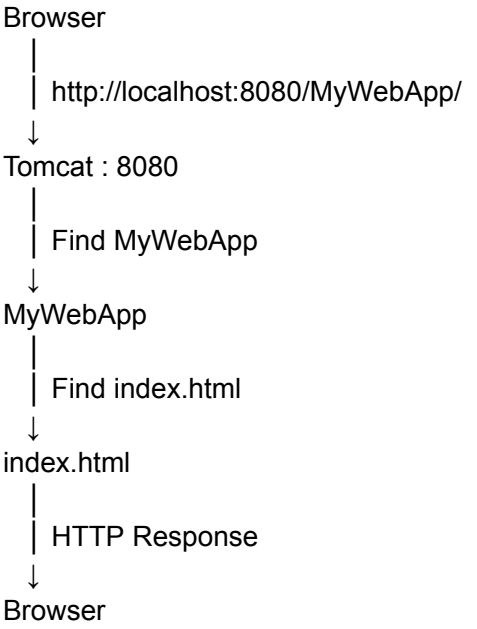
If the folder is:

webapps
└── CollegeApp
    └── index.html

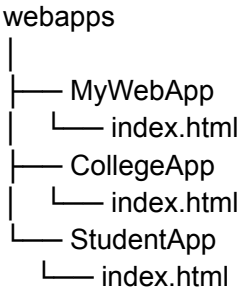
URL:
http://localhost:8080/CollegeApp/

If the folder is StudentApp:
http://localhost:8080/StudentApp/

17. Complete Request-Response Flow



18. Multiple Applications in Tomcat



One Tomcat server can run multiple web applications, each identified by its application path.

19. Important Terms

Term	Meaning
Apache Tomcat	Software used to run and serve Java web applications
Web Server	Software that receives and responds to web requests
Web Application	Application accessed through a browser
webapps	Default Tomcat directory for web applications

MyWebApp	Our web application folder
index.html	HTML welcome page
localhost	Current computer
8080	Tomcat port used in this setup
URL	Address used to access a web resource
Context Path	Path identifying a web application

20. Practical Steps

1. Download/extract Apache Tomcat.

2. Locate the Tomcat installation folder.

3. Open the bin directory.

4. Start Tomcat using startup.bat.

5. Open the webapps directory.

6. Create a folder named MyWebApp.

7. Create index.html inside MyWebApp.

8. Add HTML content.

9. Open the application URL in a browser.

10. Verify that the HTML page is displayed.

Application URL:
http://localhost:8080/MyWebApp/

21. Common Errors

Tomcat does not start: Check whether Java is configured properly and whether JAVA_HOME/JRE_HOME is available.

404 Not Found: Check that webapps → MyWebApp → index.html exists and that the URL is correct.

Wrong application name: If the folder is MyWebApp, use /MyWebApp/ in the URL.

Port already in use: Another program may already be using port 8080.

22. Viva Questions

Q. What is Apache Tomcat?
Ans. Tomcat is software used to run and serve Java web applications.

Q. Which folder contains web applications?
Ans. webapps

Q. What did we create inside webapps?
Ans. MyWebApp

Q. What is index.html?
Ans. The HTML page used as the welcome page of our basic web application.

Q. What is localhost?
Ans. It refers to the current computer.

Q. What port are we using?
Ans. 8080

Q. What is the URL of our application?
Ans. http://localhost:8080/MyWebApp/

Q. What does MyWebApp represent in the URL?
Ans. The application's context path.

Q. Can Tomcat run multiple web applications?
Ans. Yes.

Q. What happens when we enter the URL?
Ans. The browser sends a request to Tomcat; Tomcat locates the application and welcome page, and the HTML response is displayed.

23. Lab Record

Title: Installation and Configuration of Apache Tomcat and Creation of a Basic Web Application

Aim: To install Apache Tomcat and create and run a simple web application containing an HTML page.

Requirements: JDK, Apache Tomcat, VS Code, Web Browser

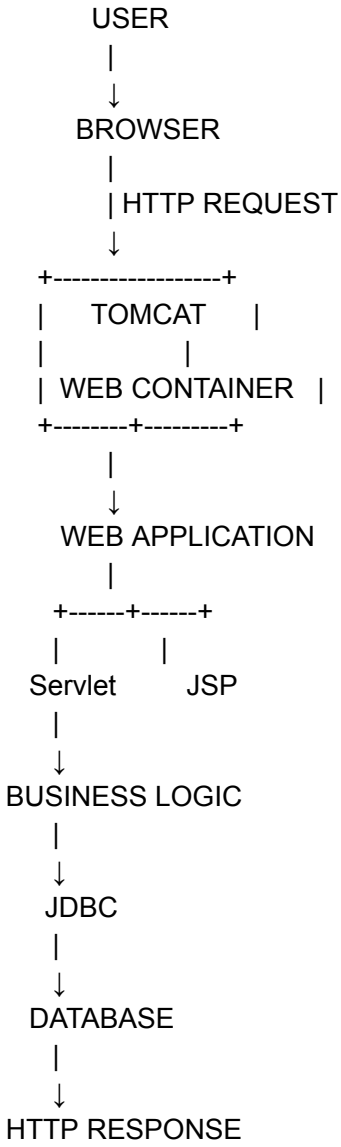
Result: A basic web application was successfully created and executed using Apache Tomcat.

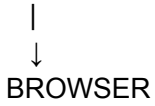
24. Key Points to Remember

Tomcat
↓
webapps

↓
MyWebApp
↓
index.html
↓
startup.bat
↓
localhost:8080
↓
/MyWebApp/
↓
Web Page

25. J2EE Architecture and Web Application Flow





26. Title

Study of J2EE Architecture and Deployment of a Simple Web Application using Apache Tomcat

27. Aim

To study J2EE architecture, enterprise application concepts, n-tier application architecture, HTTP protocol, web applications, web containers and application servers, and to deploy a simple web application using Apache Tomcat.

28. Software Required

JDK
Apache Tomcat
Web Browser
Text Editor / IDE

29. Procedure

1. Install/configure JDK.

2. Install Apache Tomcat.

3. Start Tomcat server.

4. Create a web application inside webapps.

5. Create index.html.

6. Deploy the application.

7. Access the application through the browser.

8. Observe the HTTP request and response flow.

30. URL

http://localhost:8080/MyWebApp/

31. Result

A simple web application was successfully created and deployed on Apache Tomcat.

32. Final 5-Minute Viva

Q. What is J2EE?

Answer: J2EE is a platform/collection of technologies for developing enterprise applications.

Q. What is an enterprise application?

Answer: An application designed to support business or organizational processes, often involving multiple users, business logic, databases and network communication.

Q. What is 2-tier architecture?

Answer: Client → Database

Q. What are the three tiers?

Answer: Presentation Tier, Business Tier, Data Tier.

Q. What is HTTP?

Answer: HyperText Transfer Protocol.

Q. What is an HTTP request?

Answer: A request sent by the client/browser to the server.

Q. What is an HTTP response?

Answer: A response sent by the server back to the client.

Q. What is a web application?

Answer: An application accessed through a web browser over a network.

Q. What is a web container?

Answer: A runtime environment that manages Java web components such as Servlets.

Q. What is Tomcat?

Answer: Apache Tomcat is a Java web server and Servlet container.

Q. What is port 8080?

Answer: It is the port commonly used by Tomcat in this setup.

Q. What is localhost?

Answer: It refers to the local computer.

ADDITIONAL THEORY: J2EE ARCHITECTURE & WEB APPLICATION FLOW

1. J2EE Architecture – Basic Overview

J2EE (Java 2 Platform, Enterprise Edition) is a platform/collection of technologies used for developing enterprise applications. An enterprise application commonly contains a presentation layer, business logic and a data layer.

2. N-Tier Application Architecture

A typical enterprise web application can be organized into multiple tiers. The main tiers discussed in this experiment are:

· Presentation Tier — provides the user interface and handles interaction with the user.

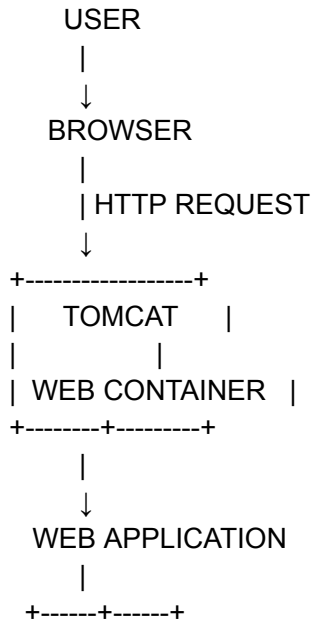
· Business Tier — contains business logic and processing.

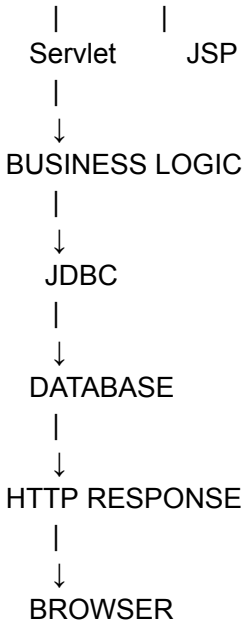
· Data Tier — manages data storage and database operations.

Typical flow:

Presentation Tier
↓
Business Tier
↓
Data Tier

3. J2EE Web Application Request-Response Flow





The browser sends an HTTP request to Tomcat. Tomcat's web container manages the web application. The web application may contain Servlets and JSP. Business logic can interact with the database through JDBC. After processing, an HTTP response is returned to the browser.

4. Enterprise Application Concepts

· Enterprise application — an application designed to support business or organizational processes.

· Web application — an application accessed through a web browser over a network.

· Web container — a runtime environment that manages Java web components such as Servlets.

· Application server — server-side software that provides an environment and services for enterprise applications.

5. HTTP Protocol

HTTP stands for HyperText Transfer Protocol. It is used for communication between a client such as a browser and a web server.

HTTP Request:
Browser → Server

HTTP Response:
Server → Browser

6. HTTP Request

An HTTP request is a request sent by the client/browser to the server for a resource or operation.

7. HTTP Response

An HTTP response is the response sent by the server back to the client/browser after processing the request.

8. Web Container and Tomcat

A web container provides the runtime environment for Java web components such as Servlets. Apache Tomcat is a Java web server and Servlet container. In this experiment, Tomcat receives the browser request and serves the deployed web application.

9. 2-Tier Architecture

Client
↓
Database

In a 2-tier architecture, the client communicates directly with the database.

10. Three-Tier Architecture

Presentation Tier
↓
Business Tier
↓
Data Tier

The presentation tier handles user interaction, the business tier handles business logic, and the data tier handles database-related operations.

11. Important Terms – Viva Revision

Question	Answer
What is J2EE?	J2EE is a platform/collection of technologies for developing enterprise applications.
What is an enterprise application?	An application designed to support business or organizational processes, often involving multiple users, business logic, databases and network communication.
What is 2-tier architecture?	Client → Database.

What are the three tiers?	Presentation Tier, Business Tier and Data Tier.
What is HTTP?	HyperText Transfer Protocol.
What is an HTTP request?	A request sent by the client/browser to the server.
What is an HTTP response?	A response sent by the server back to the client.
What is a web application?	An application accessed through a web browser over a network.
What is a web container?	A runtime environment that manages Java web components such as Servlets.
What is Tomcat?	Apache Tomcat is a Java web server and Servlet container.
What is port 8080?	The port commonly used by Tomcat in this setup.
What is localhost?	It refers to the local computer.

12. Final 5-Minute Viva

Ask students these questions orally before ending the class: What is J2EE? What is an enterprise application? What is 2-tier architecture? What are the three tiers? What is HTTP? What is an HTTP request? What is an HTTP response? What is a web application? What is a web container? What is Tomcat? What is port 8080? What is localhost?